Dataset with brain atlas A for classification tasks: for the first set of 5 small programming tasks use the Oasis dataset provided by the nilearn package, which can be downloaded using the code below. This is a dataset of structural brain scans (aka voxel-based morphometry) for male and female individuals.

```
import numpy as np
from nilearn import datasets
from nilearn.input_data import NiftiLabelsMasker
from nilearn.image import index_img
import nibabel as nib

brain_data = datasets.fetch_oasis_vbm(n_subjects=100)
yeo = datasets.fetch_atlas_schaefer_2018(n_rois=200)

masker = NiftiLabelsMasker(labels_img=yeo.maps, standardize=True,
memory='nilearn_cache')
input variables =masker.fit transform(brain data.gray matter maps)
output_variable = np.array(brain_data.ext_vars.mf == 'F',
dtype=int) #gives 1 for females and 0 for males
```

Task 1 - Use `sklearn.linear_model.LogisticRegression`(penalty='l1', C=1.0), `sklearn.linear_model.LogisticRegression`(penalty='l2', C=1.0) and `sklearn.linear_model.LogisticRegression`(penalty='elasticnet', l1_ratio=0.5, C=1.0) to fit separate logistic predictive models based on 100, 200, 300, …, 900, 1000 brain regions of interest (see 'n_rois' argument for datasets.fetch_atlas_schaefer_2018) based on the first 80% of the 100 structural brain scans to predict sex. Next, evaluate each fitted LogReg model on the unseen final 20% of the 100 structural brain scans and retrieve classification accuracy in XX.YY%. Please plot the classification accuracy on the test set (x axis=the different atlas resolutions, y axis=prediction accuracy on test set, 3 lines with 3 colors for the 3 prediction model classes with different regularization constraints).

Task 2 - Same as 1, but plot the count of exactly-zero parameters of the model fitted on the 80% training subjects (x axis=the different atlas resolutions, y axis=number of exactly-zero parameters in the entire model, 3 lines with 3 colors for the 3 prediction models with different regularization constraints).

Task 3 - The goal is to plot a 'regularization path' for a given classification problem. Use `sklearn.linear_model.LogisticRegression`(penalty='l1') to fit separate logistic predictive models based on max **1000 brain regions of interes**t (see 'n_rois' argument) based on the first 80% of the 100 structural brain scans to predict sex differences. Find a grid of 10 different values for the hyperparameter C, and **fit a different LogReg model for each of these choices of C**. Then plot the model parameter set with different colors (analogous to the regularization paths that were presented in the lecture, hyperparameter choice (C) on x axis, model parameter value on y axis), such that the extremes of this grid of increasing C values has all parameter in the model versus none of the parameters in the model. Called 'regularization path'.

Task 4 - Use `sklearn.linear_model.LogisticRegression`(penalty='l1', C=1.0) to fit separate sparse logistic predictive models based on 1000 brain regions of interest (see 'n_rois' argument) based on the first 80% of the 100 structural brain scans to predict sex differences. **This previous step should be done based on 100 bootstrapping datasets**, drawing with replacement from the 80% training data. Please plot the occurrence of non-zero model parameters across the 100 bootstrapped sparse predictive models (x axis=the 1000 parameters related to input variables, y axis=0..100 of how many times the fitted model had a given parameter set to a value != 0).

Task 5 - Same as task 4, but adding noise to input data. After drawing one of the 100 bootstrap datasets and before fitting the predictive model: add random by adding to the input data matrix the noise output from *`np.random.randn(80, 1000) * 0.1`*. Before looping over the 100 bootstrap datasets, make sure to set random seed once via *`np.random.seed(0)`*.

Dataset with brain atlas B for continous prediction tasks: For the second set of 5 small programming tasks use the Oasis dataset provided by the nilearn package, which can be downloaded using the code below. This is a dataset of structural brain scans (aka voxel-based morphometry) for male and female individuals.

```
import numpy as np
from nilearn import datasets
from nilearn.input_data import NiftiLabelsMasker
from nilearn.image import index_img
from sklearn.preprocessing import StandardScaler
import nibabel as nib

brain_data = datasets.fetch_oasis_vbm(n_subjects=100)
crad = datasets.fetch_atlas_craddock_2012()
atlas_nii = index_img(crad['scorr_mean'], 42) # this needs to be
modified for some tasks
print('%i regions found in this atlas' %
(len(np.unique(atlas_nii.get_data().ravel())) - 1))

masker = NiftiLabelsMasker(labels_img= atlas_nii, standardize=True)
input_variables = masker.fit_transform(brain_data.gray_matter_maps)
output_variable =
StandardScaler().fit_transform(brain_data.ext_vars.age[:, None])[:,
0]  # gives subject age on standard units after z-scoring
```

Note: For these tasks, instead of LogisticRegression() use the three regressors Ridge(), Lasso(), and ElasticNet(), depending on which penalty is desired. Use R2 coefficient of determination instead of classification accuracy, which can be found at *sklearn.metrics.r2_score.*

6- analogous to task 1, but predicting continuous subject age and using 43 different versions of Craddock region atlas resolutions (.scorr_mean) instead of the 10 region numbers from Dataset A. Use alpha=0.1.

7- analogous to task 2, but predicting continuous subject age and using 43 different versions of Craddock region atlas resolutions (.scorr_mean) instead of the 10 region numbers from Dataset A. Use alpha=0.1.

8- analogous to task 3, but predicting continuous subject age and using the largest (43rd, 788 input variables) of Craddock region atlas resolutions (.scorr_mean).

9- analogous to task 4, but predicting continuous subject age and using the largest (43rd, 788 input variables) of Craddock region atlas resolutions (.scorr_mean). Use alpha=0.1.

10- analogous to task 5, but predicting continuous subject age and using the largest (43rd, 788 input variables) of Craddock region atlas resolutions (.scorr_mean) instead of the 10 region numbers from Dataset A. Use alpha=0.1.